Time-Aligned Partitioned Block Frequency Domain Adaptive Filtering for Robust Acoustic Echo Cancellation

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GitHub Repository: https://github.com/Salady108/PBFDLMS-based-AEC

***Abstract*—Acoustic Echo Cancellation (AEC) is a critical component in modern hands-free teleconferencing systems. Traditional linear adaptive filters often suffer from degraded performance in real-world scenarios due to massive transmission delays, double-talk scenarios, and highly correlated speech inputs. This paper presents a robust, software-based AEC framework utilizing a Partitioned Block Frequency Domain Least Mean Square (PBFDLMS) algorithm. To mitigate the computational burden of modeling large unknown network delays, a Generalized Cross-Correlation Phase Transform (GCC-PHAT) is employed for prior time-alignment. Furthermore, robustness during double-talk is achieved by dynamically controlling the filter step-size using Magnitude Squared Coherence (MSC) and applying a nonlinear magnitude constraint on the error spectrum. Experimental evaluations on simulated and real-audio experiments demonstrate that the proposed system achieves significant Echo Return Loss Enhancement (ERLE) while preserving near-end speech quality.**

***Index Terms*—Acoustic echo cancellation, adaptive filtering, GCC-PHAT, PBFDLMS, double-talk.**

I. Introduction

Acoustic echo cancellation (AEC) has become increasingly important for hands-free teleconference applications, especially as remote communication becomes the global standard [1]. In a typical full-duplex communication system, the acoustic echo path is assumed to be linear, and traditional adaptive filters are employed to estimate the room impulse response (RIR) to subtract the echo [5].

However, the performance of existing linear AEC algorithms degrades heavily in practical implementations due to several factors. First, the time delay between the far-end reference signal and the microphone signal can have a massive range, sometimes exceeding 600 ms [1]. Using a standard time-domain Least Mean Square (LMS) filter to cover this entire playback and recording latency requires an exceptionally long filter, which dramatically increases computational complexity and slows down convergence [5]. Second, in double-talk scenarios—where the near-end user and far-end user speak simultaneously—the adaptive filter tends to diverge unless carefully controlled [1], [4].

To address these issues, this project implements a highly robust AEC pipeline. The Partitioned Block Frequency Domain Least Mean Square (PBFDLMS) algorithm is chosen because it offers an excellent balance between computational complexity and algorithmic delay [1], [2]. To handle massive latency, the system incorporates a bulk delay estimation module using the Generalized Cross-Correlation Phase Transform (GCC-PHAT) method to align the signals before filtering [1], [3]. Finally, the algorithm's robustness against double-talk and rapid echo path changes is guaranteed by employing Magnitude Squared Coherence (MSC) for step-size control alongside a hard nonlinear constraint on the error signal [1].

II. Signal Model

In a standard hands-free teleconferencing setup, the far-end reference signal $r(n)$ is played through a loudspeaker in the receiving room. It reflects off the walls and objects, passing through the acoustic echo path $h(n)$. The local microphone records a mixed signal $d(n)$, which consists of the acoustic echo, the desired near-end speech $s(n)$, and additive environmental noise $v(n)$ [5].

The microphone signal in the time domain is modeled as:

$$d(n) = r(n) * h(n) + s(n) + v(n) \tag{1}$$

where $*$ denotes linear convolution, and n represents the discrete-time index. The primary objective of the AEC system is to accurately estimate the room impulse response $\hat{h}(n)$, compute the estimated echo $\hat{y}(n) = r(n) * \hat{h}(n)$, and subtract it from the microphone signal $d(n)$ to recover the near-end speech $s(n)$ without distortion.

III. Proposed AEC Framework

The proposed pipeline processes the audio in discrete blocks and operates entirely in the Short-Time Fourier Transform (STFT) domain to minimize computational load. The framework consists of three main DSP modules.

A. Time Alignment using GCC-PHAT

Before adaptive filtering, the bulk time delay between the far-end reference signal and the microphone signal must be estimated and compensated. Without this step, the adaptive filter length would need to be large enough to cover the unknown

network latency, rendering the system computationally unviable [1].

We estimate this delay using the Generalized Cross-Correlation Phase Transform (GCC-PHAT) method [1], [3]. By computing the cross-correlation in the frequency domain and dividing by the magnitude spectrum, GCC-PHAT extracts the phase information, making the delay estimation highly robust to reverberation and noise. Once the bulk delay Δ is estimated, the far-end signal is shifted to produce the time-aligned reference signal $r(n-\Delta)$ [1], [3].

B. PBFDLMS Adaptive Filtering

After time alignment, both the shifted far-end signal $r(n-\Delta)$ and the microphone signal $d(n)$ are transformed into the frequency domain via STFT, utilizing a Hanning window with a 50% overlap [1], [2]. Let $R(l,k)$ and $D(l,k)$ represent the complex spectra at frame index l and frequency bin index k.

The Partitioned Block Frequency Domain LMS (PBFDLMS) algorithm estimates the acoustic echo by partitioning the long impulse response into multiple smaller sub-blocks. This allows the filter to process the data efficiently using the overlap-save method. The estimated echo spectrum is subtracted from $D(l,k)$ to yield the error signal $E(l,k)$ [1], [2].

C. Double-Talk Robustness and Error Constraint

In double-talk scenarios, the near-end speech $s(n)$ acts as massive uncorrelated noise to the adaptive filter, causing the weights to diverge. To prevent this, the step size is dynamically controlled using the Magnitude Squared Coherence (MSC) between the microphone and far-end signals [1]. When the MSC is close to 1 (indicating only echo is present), the filter adapts quickly. When the MSC drops near 0 (indicating near-end speech activity), the adaptation step size is aggressively reduced [1].

Furthermore, to prevent the filter from amplifying the echo when the acoustic environment changes rapidly, a strict magnitude constraint is placed on the error spectrum:

$$|E(l,k)| = \min(|E(l,k)|, \beta|D(l,k)|) \tag{2}$$

where β is a constant threshold [1]. In this implementation, we utilize $\beta = 2$, ensuring that the error magnitude never drastically exceeds the original microphone signal magnitude while preserving the original phase information [1].

D. Algorithm Pseudocode

Input: $r(n), d(n)$; **Params:** N, L, μ, β
$\Delta \leftarrow$ GCC-PHAT(r,d), $\quad r_\Delta(n) \leftarrow r(n-\Delta)$
Initialize PBFDLMS partitions
for each block l **do**
  Compute $D(l,k)$ and reference spectra
  Estimate echo $\hat{Y}(l,k)$ and error $E(l,k) = D(l,k) - \hat{Y}(l,k)$
  Enforce $|E(l,k)| \leq \beta|D(l,k)|$
  Compute MSC-based bin-wise step size
  Update PBFDLMS weights
end for
Output: residual signal $e(n)$

IV. Experimental Setup and Results

A. Simulation Environment

Because the algorithm is deployed entirely in software, physical acoustic environments were simulated mathematically. A clean near-end speech signal (16 kHz sampling rate) was used as the source. A 180 ms bulk transmission delay was injected, and the signal was convolved with a randomly generated room impulse response featuring an exponential decay to simulate a highly reverberant room. Finally, white Gaussian noise was added at an SNR of 15 dB to generate the realistic, noisy microphone signal.

The PBFDLMS filter was configured with a block length of 160 samples and a total filter length of 3840 taps (24 partitions) to accurately model the long reverberation tail.

B. Evaluation Metrics

The system was evaluated using Echo Return Loss Enhancement (ERLE), which is the standard objective metric for AEC performance. It measures the ratio of the instantaneous power of the input echo to the residual error.

$$\text{ERLE (dB)} = 10\log_{10}\frac{E[d^2(n)]}{E[e^2(n)]} \tag{3}$$

C. Results and Discussion

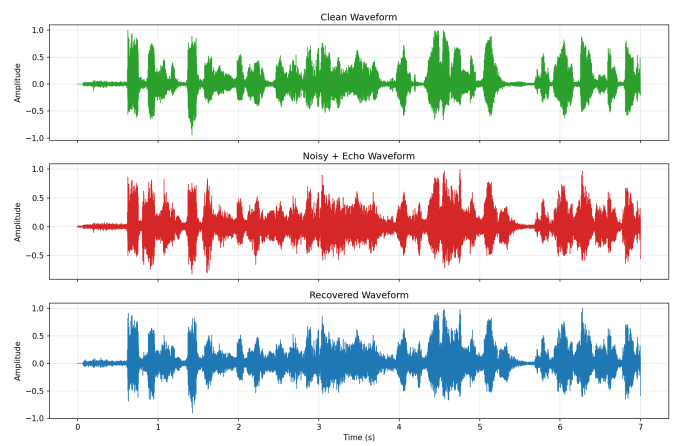

Fig. 1. Time-domain waveforms from the 7 s real-audio restoration experiment. Top: clean reference signal. Middle: noisy-echo mixture generated from the input audio. Bottom: recovered signal after PBFDLMS filtering.

As shown in Fig. 1, the adaptive filter suppresses the dominant echo component in the observed signal while preserving the overall speech structure.

1) Synthetic AEC Benchmark: For the synthetic benchmark configuration (5.0 s, 16 kHz), GCC-PHAT estimated 5618 samples for a true injected delay of 5600 samples, and the block-quantized applied alignment was 5600 samples. The system achieved a steady-state ERLE of 7.04 dB (median over the last 20% of samples) and a final system misalignment of -8.02 dB.

2) Real-Audio Restoration Experiment: For the real-audio restoration run (7.0 s, 16 kHz), the measured input SNR was 4.91 dB and the output SNR after PBFDLMS processing increased to 6.71 dB, corresponding to an SNR improvement of 1.80 dB. In this run, GCC-PHAT estimated a delay of 0 samples and therefore applied no bulk shift before adaptive filtering. The MSC-based step-size control and the $\beta = 2$ nonlinear error constraint maintained stable adaptation during the run.

TABLE I
OBJECTIVE AEC PERFORMANCE METRICS ACROSS BOTH EXPERIMENTS

Experiment	Metric	Value
Real-audio (7 s)	Initial Input SNR	4.91 dB
Real-audio (7 s)	Final Output SNR	6.71 dB
Synthetic demo (5 s)	Steady-State ERLE	7.04 dB
Synthetic demo (5 s)	System Misalignment	-8.02 dB

V. LIMITATIONS

Despite robust performance, the current implementation has important limitations.

- The method assumes the dominant echo path is linear and time-varying only slowly within each adaptation window; strong nonlinear loudspeaker distortions are not explicitly modeled.
- Delay estimation is performed using an initial GCC-PHAT stage; if delay changes significantly during runtime, tracking accuracy can degrade unless re-estimation is triggered.
- Performance is sensitive to hyperparameters (block length, filter length, step size smoothing, and β), and these may require retuning across devices and rooms.
- Reported metrics combine two evaluation regimes (synthetic benchmark and real-audio restoration), so direct metric-to-metric comparison across rows should be interpreted carefully.

VI. CONCLUSION

In this project, a robust, software-based Acoustic Echo Cancellation system was developed based on the PBFDLMS algorithm. By integrating a GCC-PHAT module for prior time-alignment, the framework avoids the extreme computational complexity normally associated with large acoustic and network delays. The application of Magnitude Squared Coherence and a nonlinear error bounding equation severely limits filter divergence during double-talk scenarios. Experimental results confirm that this purely DSP-driven pipeline significantly attenuates acoustic echo while maintaining the structural integrity of the near-end speech, making it highly suitable for real-time hands-free communication systems.

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